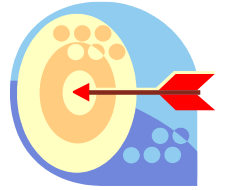


Lesson 5: Using the Graphical Process Modeler (GPM)

Lesson Objectives



This lesson provides instructions for using the Graphical Process Modeler (GPM) to create an executable business process and is designed to enable you to:

- Define a business process and the life cycle of the business process
- Create a business process
- Implement a business process
- Check in a business process model
- Enable a business process model
- Manually run a business process model
- Check out a business process
- Modify a business process
- Check in a modified business process
- Disable a business process

Business Process

What is a Business Process?

Business processes are like road maps. Just as you use a road map to determine the best route through the countryside, Sterling B2B Integrator uses business processes to find the best route through critical business systems, such as purchasing, inventory status , and invoicing.

This section provides an overview of how Sterling B2B Integrator works. It will follow Sterling B2B Integrator through:

- Defining a business process with services
- Compiling a business process
- Check In a business process
- Starting a business process
- Archiving a business process
- Restoring a business process
- Deleting a business process

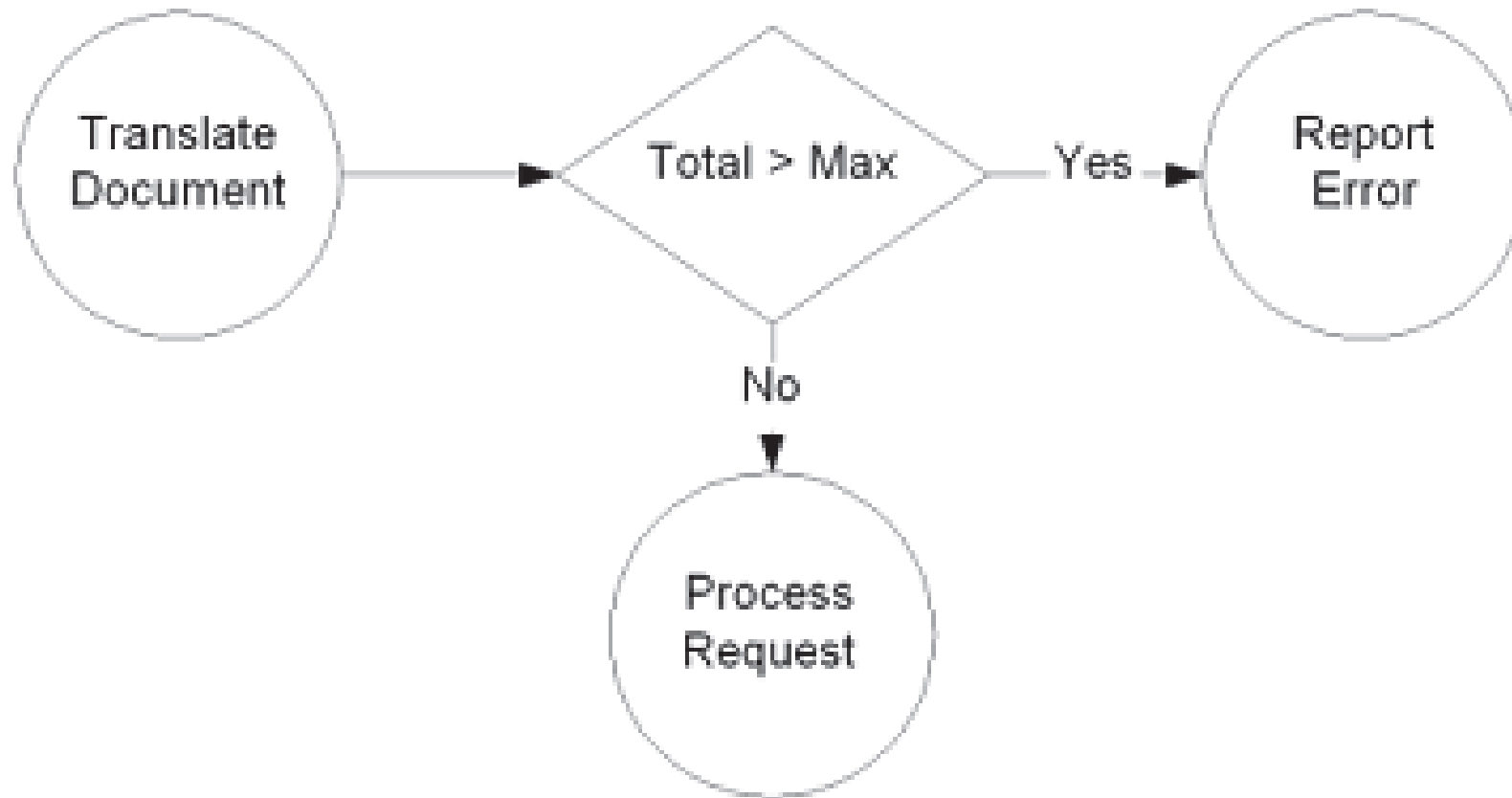
Life Cycle of a Business Process

This section provides an overview of how a business process flows through Sterling B2B Integrator:

- Analyzing a business process
- Defining a business process with services
- Compiling a business process
- Check In a business process
- Starting a business process
- Archiving a business process
- Restoring a business process
- Deleting a business process

Defining a Business Process Model

Defining a business process might look like the following diagram. The circles represent activities and the diamond represents a decision point.



Business Process Nodes

Nodes in a business process model

Nodes in a business process model are the graphical representation of the steps the business process goes through as it runs and processes the data involved.



Services

Services

- Internal services are located inside Sterling B2B Integrator; they accept parameters and produce results, but they do not interact with any system outside Sterling B2B Integrator.
- Input services receive data from an outside system.
- Output services send data to an outside system.

Adapters

- Input and output services are known collectively as adapters.
- Adapters interact with the “world” outside Sterling B2B Integrator.

Compiling a Business Process

After you create a business process model, you must validate and compile the process before you can run it. Validating ensures all activities in the business process model are properly configured. Compiling the business process allows Sterling B2B Integrator to determine the “Start” node ahead of time so business processes are more easily instantiated and run. Since the start pointer for the next activity (node) is stored in the current activity (node) information, compiling the business process also eliminates repeated parsing of XML and reduces the number of queries to the database.

Checking in a Business Process

Before you can use a business process within Sterling B2B Integrator, you must first check it into the system. Each time you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

Starting a Business Process

Business processes can be started either **manually or automated** by the system.

Automated can be:

- **Scheduled-** Set to run a specific times or intervals
- **Invoked-** Called or started by another business process.
- **Bootstrapped-** Similar to Invoked, but called from an adapter or event instead of a business process. Such as an adapter monitoring a file folder and kicking off a business process when a file is found. Dynamic bootstrapping is the selection of the appropriate business process to run when data enters. Input data must enter Sterling B2B Integrator by way of an input adapter. This adapter takes the received data, puts the data and any metadata it finds into an Initial Business Process Context (IBPC), and begins bootstrapping. A business process definition is selected based on the data and/or metadata and instantiated as needed.

Administering Business Process Instances

Archiving a Business Process - Archiving your business process data not only protects your critical data but also conserves database storage, improving the efficiency of Sterling B2B Integrator.

Restoring a Business Process - After you have archived business process data to an offline location, you can restore that data and view it in Sterling B2B Integrator.

Deleting a Business Process

Deleting a Business Process - In Sterling B2B Integrator, you can delete individual business processes (except business processes used by Sterling B2B Integrator). Some reasons you might delete a business process are:

- Business activities have changed and, as a result, a business process can no longer accomplish the activities.
- You need to remove versions of a business process to reduce the database size.

Important!!

A business process or a business process version can not be deleted if there is an active instance in the database. If a business process runs and has a lifespan of 2 days. That instance record will remain in the database for 2 days and the business process can not be deleted until that instance has been purged.

Creating a Business Process

Creating a Business Process - After analyzing your business needs, determining your business processes, and documenting the results, you must create business process models that represent those business processes. You can then implement and run the business processes with Sterling B2B Integrator.

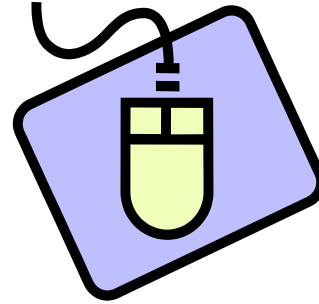
Important!!

Because a business process must be validated to run successfully in Sterling B2B Integrator, you must ensure that you validate the BPML code in each business process before you add the process to Sterling B2B Integrator. The sections in this lesson include steps to validate a business process.

Implementing a Business Process Using the GPM

You can implement a business process in Sterling B2B Integrator using the Graphical Process Modeler (GPM) by following the steps listed below as appropriate:

1. Create a business process model, including:
 - Configuring additional parameters
 - Adding sub flows
 - Assigning rules and conditions
 - Assigning name-value pairs that represent data to locate in process data
 - Configuring the Correlation service to generate correlations for business processes and document exchange
2. Validate and save the business process model document.
3. Add the business process to Sterling B2B Integrator.



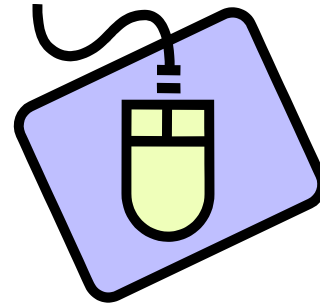
Exercise(s) 5-1 and 5-2: Moving Activities to the Workspace And Connecting Activities

Your turn!

Pause this presentation, read the Scenario then follow the instructions to Move Activities to the Workspace

Sub-Flows

- When you create a subflow, you create a layer in the business process model when viewed in the GPM. You can expand the view to work with the components of the subflow by using the Layers tab in the Navigation pane.
- While a subflow layer provides more design space in the GPM, it does not create a separately-saved business process model. Nor does it create a child business process that is called from a parent.
- Subflows are only available in the GPM



Exercise 5-3: Adding Sub flows

Your turn!

Pause this presentation, read the Scenario then follow the instructions to Adding Sub flows

Assigning Rules and Conditions

When you have a decision point in a business process you must use a rule that describes the decision. And provide the conditions where a true or false choice can be made.

When running a business process, you must use XPath to describe rules and conditions for the rule.

Some examples of using XPath expressions in a rule are:

- Identify particular elements (or nodes) and their positions in both the input and output documents.
- Identify particular elements in the input document for copying into an output document or for further processing.
- Define any constraints of name-value pairs for elements.
- Calculate values that depend on other values.

You can then assign rules that represent the decision to complete activities in the business process.

Assigning Rules and Conditions

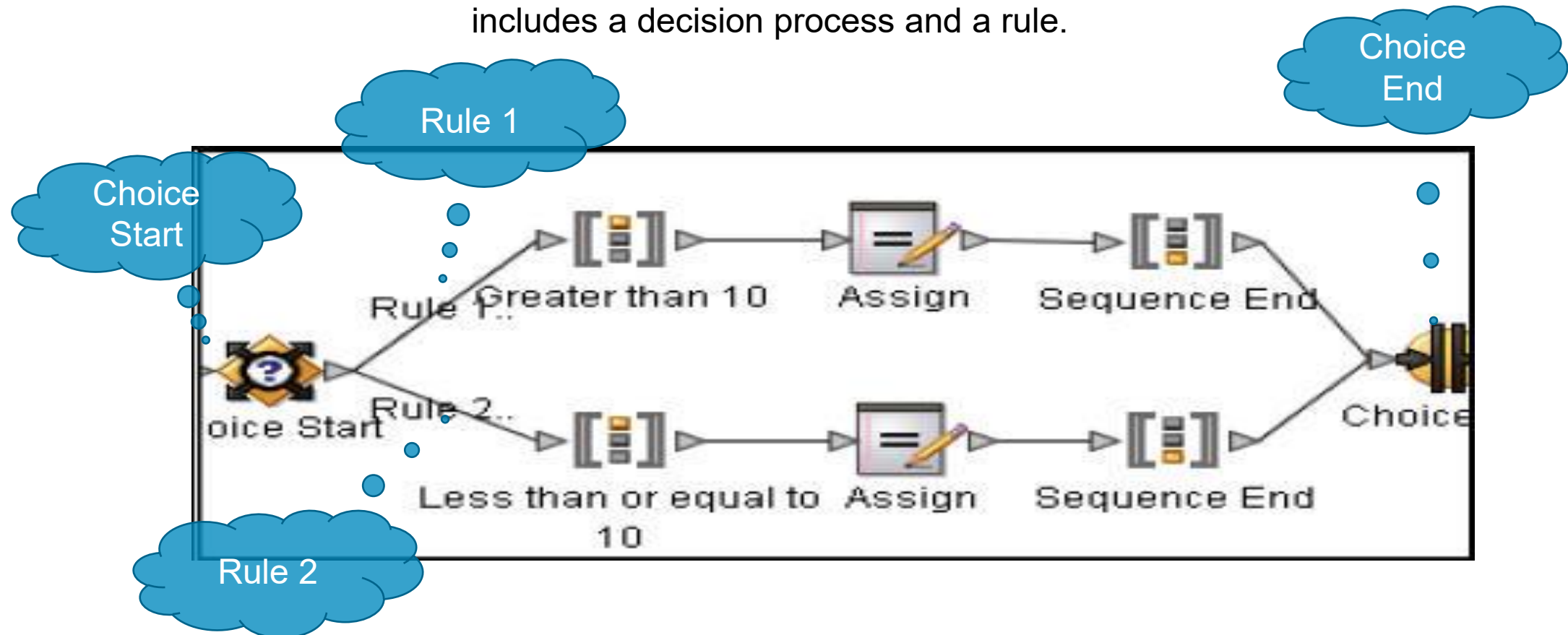
Example of Rules using simple syntax

Name	Rule
rule1	X>10
rule2	X<=10
rule3	X<10

- Rule1 – X is greater than 10.
- Rule2 – X is less than or equal to 10
- Rule3 – X is less than 10

Assigning Rules and Conditions

Example of a Business Process that includes a decision process and a rule.



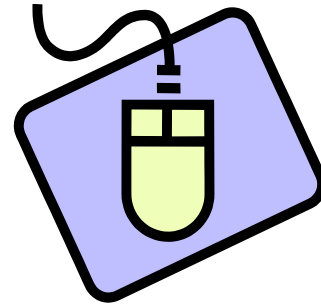
Tools for Building Rules

Rule Manager

- The Rule Manager is a tool in the GPM that is used to create rule conditions that will be evaluated in your business process.
- A rule is written in XPath and evaluates XML data in Process Data.

Edge Editor

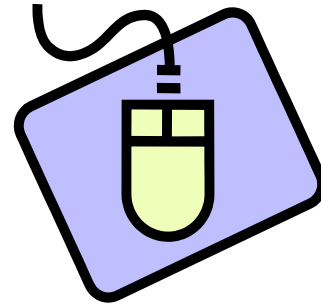
- The Edge Editor is used to assign the rules you created to a choice in your bp.
- Bring up the Edge Editor by double clicking a line coming off of your Choice Start.
- You must have a rule already created because the Edge Editor only allows you to select a rule and then say if it is true or false. This is why the name of the selected rule appears on the line.



Exercise(s) 5-4 and 5-5: Using Rules and Assigning Rules

Your turn!

Pause this presentation, read the Scenario then follow the instructions to Using Rules and Assigning Rules.



**Exercise 5-6: Viewing and Configuring the Different
Element Properties
And
Exercise 5-7: Saving and validating a business process**

Your turn!

Pause this presentation, read the Scenarios then follow the instructions to Using Rules and Assigning Rules. And Saving and Validating a Business Process.

Implementing a Business Process

Implementing a Business Process

After creating or modifying a business process model, you must validate, check in, and test it before running it in Sterling B2B Integrator. Use a test environment for checking in and testing your processes before you run them in your production environment.

The first time you check in a business process, you create a business process definition—a uniquely named copy of the process model to be stored in Sterling B2B Integrator. The business process model stored in Sterling B2B Integrator is independent of any .bp file of the process that you may have saved in a local directory.

At any time, you can modify an existing process model and check in a new version to Sterling B2B Integrator, or save the modified copy of the process in Sterling B2B Integrator as a new process altogether. Version management is an important part of business process maintenance.

Implementing a Business Process (continued)

You can check in a business process model from within the GPM interface or through the Sterling B2B Integrator interface.

Both the Sterling B2B Integrator interface and the GPM provide a Source Manager feature you use to check in a business process. The two Source Managers differ slightly, but both enable you check in new business process definitions and new versions of previously checked-in process models. Both display the current list of business process models and versions.

Each time you check in a business process, you create a version of the business process, either the first version of a new process model, or a new version of a previously checked-in business process model. Sterling B2B Integrator saves each checked in version so that you retain a copy of every iteration of a process model that you check in. You cannot check in a version of a business process that is checked out and locked (protected) by another user.

The check in procedure differs depending on your goal and the application you are using.

Implementing a Business Process (continued)

Determining Processing Options During Check-in -

During the check-in procedure, Sterling B2B Integrator prompts you to select a variety of processing options for a business process model. These options are related to archiving, document storage and tracking, persistence levels and other instructions related to your business needs. The settings you choose may affect system performance. Before you check in a business process model, have an idea of what the appropriate selections are for your installation.

Descriptions of the Processing Options are explained in detail in your student manual.

Implementing a Business Process (continued)

Enabling Document Tracking -

Enabling Document Tracking makes the document activity for this business process available from the Advanced Search menu. Access Document Tracking from **Business Process > Monitor > Advanced Search > Documents**.

Setting onFault Processing –

Select this option to enable the on-fault activity that is specified in the process to immediately execute in the event of a system error

Setting the Persistence Level -

The persistence level options enable you to specify, at the business process level, the types of data to retain in the database for the process whenever it runs. Persisted data can be crucial to process recovery in the event of errors and is useful for monitoring and tracking activities, including generating status reports.

Implementing a Business Process (continued)

Setting the Persistence Level (Continued)

Persistence can also be set globally (at the system level) for all processes; by default this is set to FULL persistence, although it may have been customized for your installation. If your installation uses the default FULL setting for the system, you generally do not need to change persistence at the business process level unless you are seeking performance improvements. For the process model you are checking in, your selection here overrides the global persistence level set in the properties file.

Important!! Two exceptions may affect your persistence selection. First, remember that some services support only a limited selection of persistence settings, regardless of the level of persistence you assign to the process model. Second, any persistence overrides configured in the BPML for steps in the process model override the setting you make here for the specified step or steps.

Implementing a Business Process (continued)

Choosing Document Storage -

Document storage type is the storage type you want for documents that process when the business process runs. The levels are:

- **Database** - Store documents in the database.
- **File System** - Store documents in the file system. This option may be best for large documents. Using this method, you must handle archiving and purging the documents from your file system.
- **System Default** - Store documents in the file system or database, according to how you configured archiving and purging in Sterling B2B Integrator.
- **Inherited** - If this is a sub-process, use the same storage type indicated for the parent business process.

When you check in a business process, Sterling B2B Integrator gives you the opportunity to select the location for storing the processed document.

Implementing a Business Process (continued)

Setting the Process Instance Life-Span -

When you check in a business process, Sterling B2B Integrator gives you the opportunity to set the Life Span of specific instances generated by that business process. The Life Span setting tells Sterling B2B Integrator how long the instance should remain in the active system.

Setting the Expired Characteristics of a Process Instance -

When you check in a business process, Sterling B2B Integrator gives you the opportunity to determine if the specific instances of the business process is to be Archived or Purged.

- **Archive** moves instances of business processes that have been run and any associated data from the live system to a storage area. Once in the storage area the business process instance (and data) is available to be moved to offline media for storage.
- **Purge** permanently removes business process instances that have run and any associated data. Archive and purge help to limit the amount of information kept in the live system. It is important to note, data that must be kept long-term should be archived and stored in a safe media.

Implementing a Business Process (continued)

Enabling a Business Process -

You must enable a business process that is checked in to the system before it can be available to run. Enabling a process model tells the system that this process is available to run. You cannot enable a business process unless it was validated at check in. Business processes checked in through the Administration UI are automatically validated.

Creating Permissions -

Select this option to have a permission selection created for this business process model automatically. The permission can be assigned to user accounts to enable or restrict user access to running the process. Permissions are optional. If none are assigned, the process is available for all users.

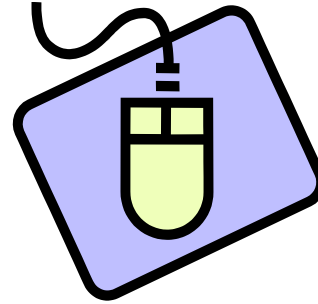
Implementing a Business Process (continued)

Setting the Default Version of a Business Process -

When you check a process model into the system, you may also set it as the default process. The default process is the process version that Sterling B2B Integrator uses when running a request of a new process instance that does not have explicit version information. We'll discuss versions of business processes later in this lesson.

Disabling a Business Process -

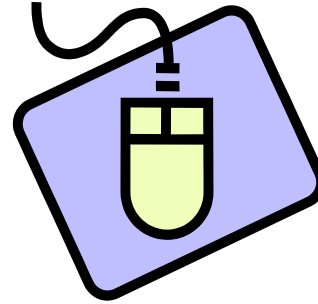
Only business processes set as the default version can run. A business process that has been checked in to the system can be disabled, or made not available to run. When a business process version is disabled, the system will not start new instances of the process.



**Exercise 5–8: Checking a Business Process using GPM
And
Exercise 5–9 Checking in a Business Process using
Business Process Manager**

Your Turn!

Pause this presentation, read the Scenario(s) then follow the instructions on Checking a Business Process Using GPM and Checking-in a Business Process using the Business Process Manager.



Exercise 5–10 Manually Running a Business Process and Exercise 5–11 Obtaining Results

Your turn!

Pause this presentation, read the Scenario(s) then follow the instructions on Manually Running a Business Process and Obtaining Results

Modifying a Business Process

Checking Out a Business Process

When you need to modify or reuse a business process that is currently checked in to Sterling B2B Integrator and enabled for modification, you can check out the business process from Sterling B2B Integrator. The check-out process enables you to save the business process to your client computer. You can then modify the business process document to create new versions of a business process or a new business process. If you have logged in to Sterling B2B Integrator using Source Manager in the GPM, you can check out a business process using Source Manager. The Source Manager also enables you to check out one or many business processes at the same time.

You may want to check out a business process when:

- A business analyst creates or modifies the business processes and has
- limited access to the host.
- An employee modifies the business processes from a remote location.
- A business process exists on a test environment and is ready for use in the production environment.

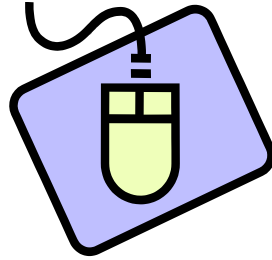
Note: If the selected business process is enabled, it remains enabled until it is replaced by a newer version.

Modifying a Business Process (continued)

Making Changes to an Existing Business Process Model –

To make changes to a process model that has been checked into Sterling B2B Integrator, you first need to check out the process. This locks the process so that no other users will make simultaneous changes. The checked out version of a process model remains locked to other users until that user checks in the process.

The GPM is the preferred method of making changes to a business process model, although advanced users may prefer to make changes using BPML. When you choose to work in the BPML, and then reopen the process model in the GPM, it will open the process using the default layout options. Since the default layout may appear differently than your specified layout, we recommend that you work in the process modeler.



Exercise 5–12 Editing a Business Process Using the GPM
And
Exercise 5–13 Editing a Business Process Using BPML
And
Exercise 5–14 Checking out the Business Process with the Edited BPML

Your turn!

Pause this presentation, read the Scenario(s) then follow the instructions on Exercises 5-12 thru 5-14

Version Control

Version Control

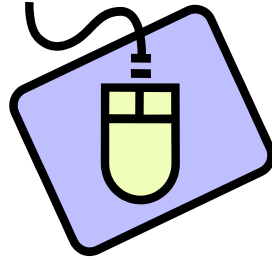
Sterling B2B Integrator maintains a record of each business process, when it is checked in, when it is checked out, and when it is modified. This simple process keeps two users from inadvertently overwriting each other's modified processes.

As each business process definition is checked into the system, a versioning record begins. Furthermore, as a different version of a process is checked in, you specify the default version of that process. This is how Sterling B2B Integrator knows which version of a process to run when new data enters the system.

Important!!!

You must also validate and enable the business process upon check-in for that version to become the default.

It is recommended that you use the description for describing the different versions.



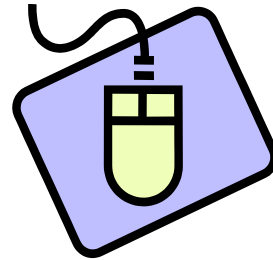
**Exercise 5–15 Viewing a Business Process Definition
Version from within the GPM Source Manager
And
Exercise 5-16 Viewing a Business Process Definition
Version from within Sterling B2B Integrator**

Your turn!

Pause this presentation, read the Scenario(s), then follow the instructions on
Viewing a Business Process Definition Version from within the GPM Source
Manager
And
Viewing a Business Process Definition Version from within Sterling B2B
Integrator



Please pause this presentation and complete the Quiz.



Exercise 5–17 Completing the Process

Your turn!

Pause this presentation, read the Scenario then follow the instructions on Completing the Process

Lesson Review



Congratulations !!!!!

Completed Lesson Objectives -

This lesson was designed to enable you to:

- Create a business process
- Implement a business process
- Check in a business process model
- Enable a business process model
- Manually run a business process model
- Modify a business process
- Check in a modified business process
- Disable a business process model